

LAPORAN PRAKTIKUM IMPLEMENTASI IPv4 DI VLAN



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POLITEKNIK ELEKTRONIKA NEGERI SURABAYA

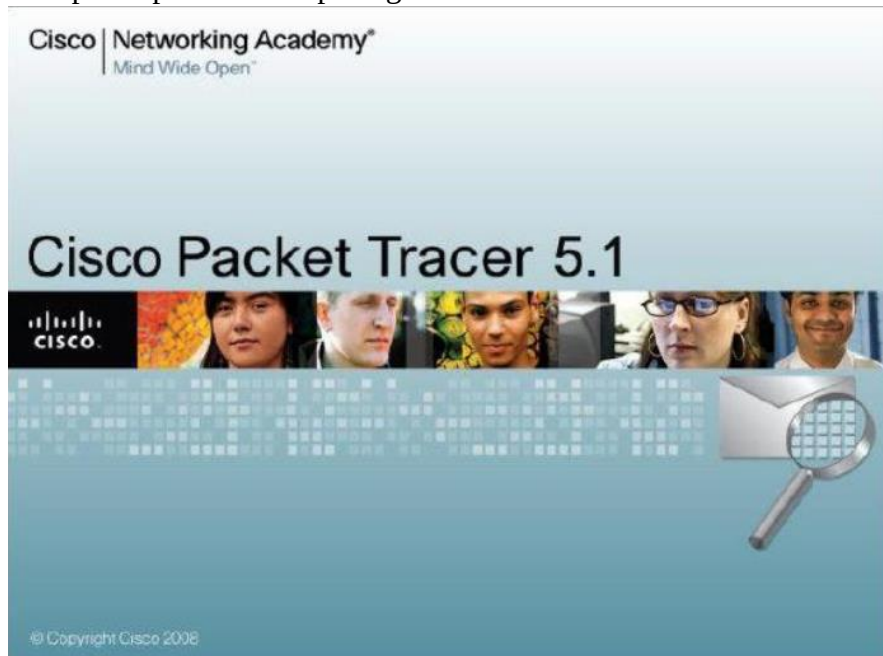
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IMPLEMENTASI IPv4 DI VLAN

A. Dasar teori

1. Cisco Packet Tracer

Cisco Packet Tracer adalah tools e-learning yang dibuat oleh Cisco yang akan mensimulasi cara kerja suatu jaringan berdasarkan topologi dan konfigurasi yang diberlakukan oleh penggunaannya persis seperti aslinya. Versi perangkat lunak Cisco Packet Tracer yang digunakan penulis saat membuat tulisan ini adalah versi 5.1 seperti yang terlihat pada splash screen pada gambar .



2. Virtual LAN (VLAN)

VLAN (Virtual Local Area Network) adalah sebuah metode atau konsep yang menggabungkan beberapa broadcast domain menjadi satu collision domain. Penerapan konfigurasi VLAN dapat dilakukan pada manageable switch. Port dengan VLAN ID yang sama dikatakan dalam satu broadcast domain. Jika VLAN ID nya berbeda, maka broadcast domainnya pun berbeda dan tidak dapat berkomunikasi. VLAN bekerja di IEEE 802.1Q.

Vlan merupakan pembagian atau segmentasi antar perangkat yang terhubung dengan salah satu switch dengan tujuan memisahkan antara satu ruangan(segmentasi) satu dengan yang lain walaupun menggunakan ip address yang masih dalam satu jaringan. Cara mengatur vlan yaitu dengan membagi setiap port yang ada pada switch ke dalam segmentasi-segmentasi yang berbeda. Pembagian tersebut dilakukan dengan memberikan nama vlan yang berbeda-beda pada masing-masing port yang terhubung dengan komputer atau perangkat lain

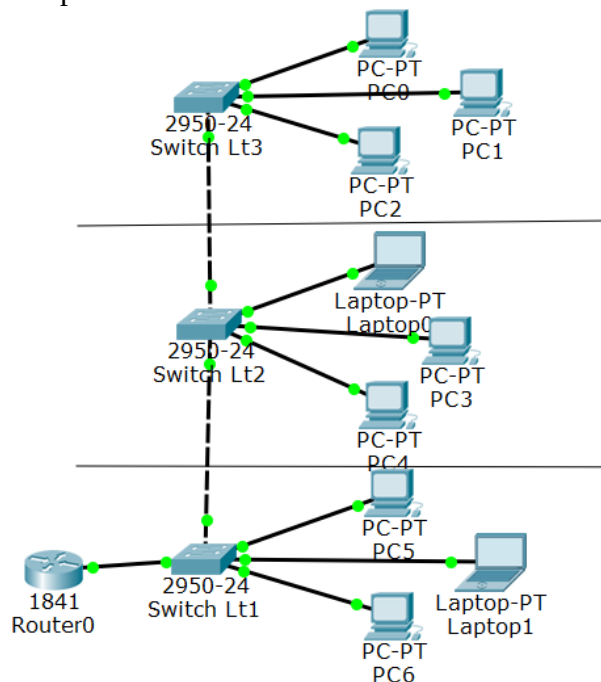
Sebuah VLAN membolehkan banyak Virtual LANs berdampingan dalam sebuah fisik LAN (switch). Artinya jika ada dua mesin yang terhubung dalam switch yang sama tidak dapat mengirim Ethernet frames ke mesin lain meskipun dalam satu kabel yang sama. Jika dibutuhkan untuk komunikasi, maka sebuah router harus ditempatkan di antara dua VLAN tersebut untuk memforward paket, seperti jika ada dua LAN yang secara fisik terpisah.

3. Tipe-tipe VLAN :

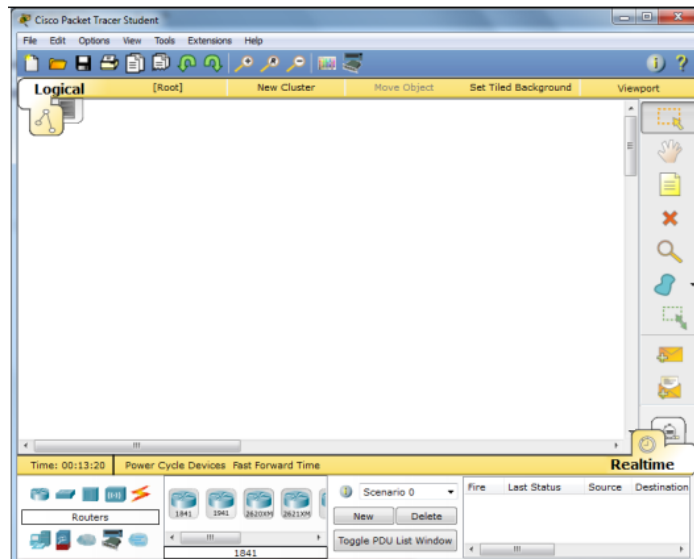
- a. Berdasarkan port Keanggotaan Pada VLAN dapat didasarkan pada port yang digunakan oleh VLAN tersebut. Sebagai contoh pada Switch dengan 4 port.
- b. Berdasarkan MAC Address Keanggotaan suatu VLAN di dasarkan pada MAC Address dari setiap workstation komputer yang dimiliki oleh user. Kelebihannya apabila user berpindah-pindah, dia akan tetap terkonfigurasi sebagai anggota dari VLAN tersebut.
- c. Berdasarkan tipe protocol yang digunakan Keanggotaan VLAN juga bisa bedasarkan protocol yang digunakan. Keuntungannya adalah seorang user tidak perlu mengkonfigurasi ulang alamatnya di jaringan apabila berpindah tempat.
- d. Berdasarkan aplikasi atau kombinasi Sangat memungkinkan untuk menentukan suatu VLAN berdasarkan aplikasi yang dijalankan, atau kombinasi dari semua tipe untuk diterapkan pada suatu jaringan. Tipe koneksi dari VLAN dapat dibagi menjadi 3, yaitu: 1. Trunk Link 2. Access Link 3. Hybrid Link
- e. 5. Berdasarkan Alamat Subnet IP Suatu subnet IP Address pada suatu jaringan routing pada jaringan juga dapat digunakan untuk mengklasifikasi suatu VLAN tabel IP Subnet dengan VLAN. Keuntungannya adalah seorang user tidak perlu untuk mengkonfigurasi ulang alamatnya di jaringan apabila berpindah tempat.

B. PROSEDUR PERCOBAAN

1. Praktikum ini berisi cara konfigurasi perangkat yang digunakan dalam jaringan komputer. Dengan contoh pada sebuah perusahaan yang memiliki 100 karyawan dengan 15 HRD, 25 Finance dan 50 marketing. Perusahaan tersebut memiliki 3 lantai dimana setiap lantai terbagi atas.. Pada praktikum kali ini buatlah konfigurasi jaringan gedung 3 lantai, dimana setiap lantai memiliki 1 switch dan 1 router pada lantai 1. Setiap switch terhubung dengan 3 PC, masing-masing PC memiliki . buatlah konfigurasi untuk membangun jaringan komputer menggunakan perangkat-perangkat yang sederhana menggunakan simulasi packet tracer.



2. Pastikan Cisco Packet Tracer sudah terinstall di komputer yang dipakai.
3. Buka Packet Tracer pada menu Start → All Programs → Cisco Packet Tracer → Cisco Packet Tracer.



4. Langkah pertama yang harus dilakukan yaitu dengan membuat terlebih dahulu desain jaringan pada aplikasi tracer dengan dengan 1 router, 3 switch dan 9 PC. Sorot Device-Type Selection Box, pilih End Devices.



5. Untuk menghubungkan semua komponen, pilih jenis kabel Cooper. Cooper straight through untuk menghubungkan router ke switch, sedangkan untuk menghubungkan switch ke switch menggunakan Cooper Cross-Over.



6. Untuk menandakan bahwa device/komputer terhubung, ditandai dengan warna hijau oada ujung kabel yang menyatakan bahwa devicece tersebut dalam keadaan connect sedangkan merah menandakan belum terhubung (karena belum dikonfigurasi/ salah menggunakan kabel). Tana merah yang menghungungkan seperti yang terlihat pada gambar diatas bukan berarti disebabkan penggunaan kabel yang salah melainkan karena belum dilakukan konfigurasi routernya.
7. Komponen yang terpasang tersebut belum mempunyai IP address. Untuk melakukan setting IP address dapat dilakukan pada properties setiap device dengan cara klik device tersebut, kemudian pilih tab panel bernama terminal. Setting router terlebih dahulu secara manual dengan melihat perintah konfigurasi dibawah, ganti nama router menjadi gateway.

Device/computer	IP Address	Fa 0/0	Fa 0/1	Fa 0/2	Fa 0/3	Fa 0/4	Fa0/5	Default Gateway	Subnet Mask
Router 0		-	192.168.0.1	192.168.1.1	192.16.8.2.1			-	255.255.255.240
Switch It 1			Vlan 10	Vlan 20		Vlan 30			

Switch lt 2					Vlan 10	Vlan 20	Vlan 30		
Switch lt 3					Vlan 10	Vlan 20	Vlan 30		
PC5 Vlan 10 lt 1 DHCP	192.168.0.3	-	-	-	-	-	-	192.168.0.1	255.255.255.240
Laptop1 Vlan 20 lt 1 DHCP	192.168.1.7	-	-	-	-	-	-	192.168.1.1	255.255.255.224
PC6 Vlan 30 lt 1 DHCP	192.168.2.1	-	-	-	-	-	-	192.168.2.1	255.255.255.128
Laptop0 Vlan 10 lt 2 DHCP	192.168.0.5	-	-	-	-	-	-	192.168.0.1	255.255.255.240
PC3 Vlan 20 lt 2 DHCP	192.168.1.8	-	-	-	-	-	-	192.168.1.1	255.255.255.224
PC4 Vlan 30 lt 2 DHCP	192.168.2.1	-	-	-	-	-	-	192.168.2.1	255.255.255.128
PC0 Vlan 10 lt 3 DHCP	192.168.0.4	-	-	-	-	-	-	192.168.0.1	255.255.255.240
PC1 Vlan 20 lt 3 DHCP	192.168.1.6	-	-	-	-	-	-	192.168.1.1	255.255.255.224
PC2 Vlan 30 lt 3 DHCP	192.168.2.1	-	-	-	-	-	-	192.168.2.1	255.255.255.128

a. Konfigurasi Router

Konfigurasi Router menggunakan IOS Command Line Interface (CLI). Untuk konfigurasi disini kita akan membuat 3 buah VLAN yaitu VLAN10, VLAN20, VLAN30.

```

Router>
Router>enable
Router#ls
Translating "ls"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Gateway
Gateway(config)#interface fastEthernet 0/0
Gateway(config-if)#no shutdown

Gateway(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Gateway(config-if)#exit
Gateway(config)#exit
Gateway#
%SYS-5-CONFIG_I: Configured from console by console

Gateway#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Gateway(config)#interface fastEthernet 0/0.1
Gateway(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.1, changed state to up

Gateway(config-subif)#no shutdown
Gateway(config-subif)#ip add
% Incomplete command.
Gateway(config-subif)#en
Gateway(config-subif)#encapsulation
% Incomplete command.
Gateway(config-subif)#no shutdown
Gateway(config-subif)#?

```

```

    arp          Set arp type (arpa, probe, snap) or timeout
    bandwidth    Set bandwidth informational parameter
    delay        Specify interface throughput delay
    description   Interface specific description
    encapsulation Set encapsulation type for an interface
    exit         Exit from interface configuration mode
    ip           Interface Internet Protocol config commands
    ipv6         IPv6 interface subcommands
    mtu          Set the interface Maximum Transmission Unit (MTU)
    no           Negate a command or set its defaults
    shutdown     Shutdown the selected interface
    standby      HSRP interface configuration commands
Gateway(config-subif)#
Gateway(config-subif)#?
    arp          Set arp type (arpa, probe, snap) or timeout
    bandwidth    Set bandwidth informational parameter
    delay        Specify interface throughput delay
    description   Interface specific description
    encapsulation Set encapsulation type for an interface
    exit         Exit from interface configuration mode
    ip           Interface Internet Protocol config commands
    ipv6         IPv6 interface subcommands
    mtu          Set the interface Maximum Transmission Unit (MTU)
    no           Negate a command or set its defaults
    shutdown     Shutdown the selected interface
    standby      HSRP interface configuration commands
Gateway(config-subif)#ip add
% Incomplete command.
Gateway(config-subif)#encapsulation dot1Q
% Incomplete command.
Gateway(config-subif)#encapsulation dot1Q 10
Gateway(config-subif)#ip add
% Incomplete command.
Gateway(config-subif)#ip address 192.168.0.1 255.255.255.240
Gateway(config-subif)#no shutdown
Gateway(config-subif)#exit
Gateway(config)#interface fastEthernet 0/0.2
Gateway(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up

Gateway(config-subif)#en
Gateway(config-subif)#encapsulation dot
Gateway(config-subif)#encapsulation dot1Q 20
Gateway(config-subif)#ip address 192.168.1.1 255.255.255.224
Gateway(config-subif)#no shu
Gateway(config-subif)#no shutdown
Gateway(config-subif)#exit
Gateway(config)#interface fastEthernet 0/0.3
Gateway(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state to up

Gateway(config-subif)#enc
Gateway(config-subif)#encapsulation
% Incomplete command.
Gateway(config-subif)#encapsulation dot
Gateway(config-subif)#encapsulation dot1Q 30
Gateway(config-subif)#ip add
Gateway(config-subif)#ip address 192.168.2.1 255.255.255.128
Gateway(config-subif)#no shu
Gateway(config-subif)#no shutdown
Gateway(config-subif)#exit
Gateway(config)#
Gateway>enable
Gateway#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Gateway(config)#ip dhcp pool vlan
Gateway(config)#ip dhcp pool vlan10

```

```

Gateway(dhcp-config)#?
  default-router  Default routers
  dns-server      Set name server
  exit            Exit from DHCP pool configuration mode
  network         Network number and mas
  no              Negate a command or set its defaults
  option          Raw DHCP options
Gateway(dhcp-config)#default-router 192.168.0.1
Gateway(dhcp-config)#network 192.168.0.0 255.255.255.240
Gateway(dhcp-config)#exit
Gateway(config)#ip dhcp pool vlan20
Gateway(dhcp-config)#default-router 192.168.1.1
Gateway(dhcp-config)#network 192.168.1.0 255.255.255.224
Gateway(dhcp-config)#exit
Gateway(config)#ip dhcp pool vlan30
Gateway(dhcp-config)#default-router 192.168.2.1
Gateway(dhcp-config)#network 192.168.2.0 255.255.255.128
Gateway(dhcp-config)#exit
Gateway(config)#ip dh
Gateway(config)#ip dhcp ex
Gateway(config)#ip dhcp excluded-address 192.168.0.1 192.168.0.2
Gateway(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.5
Gateway(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.10
Gateway(config)#%DHCPD-4-PING_CONFLICT: DHCP address conflict: server pinged
192.168.1.7.

```

b. Konfigurasi Switch Lantai 1

Konfigurasi Router menggunakan IOS Command Line Interface (CLI). Untuk konfigurasi disini kita akan membuat 3 buah VLAN yang diberinama VLAN10 = HRD, VLAN20 = Finance, VLAN30 = Marketing.

```

Switch_Ltl1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch_Ltl1(config)#vlan 10
Switch_Ltl1(config-vlan)#vlan 20
Switch_Ltl1(config-vlan)#vlan 30
Switch_Ltl1(config-vlan)#exit
Switch_Ltl1(config)#exit
Switch_Ltl1#
%SYS-5-CONFIG_I: Configured from console by console

Switch_Ltl1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch_Ltl1(config)#vlan 10
Switch_Ltl1(config-vlan)#name HRD
Switch_Ltl1(config-vlan)#exit
Switch_Ltl1(config)#vlan 20
Switch_Ltl1(config-vlan)#name Finance
Switch_Ltl1(config-vlan)#exit
Switch_Ltl1(config)#vlan 30
Switch_Ltl1(config-vlan)#name Marketing
Switch_Ltl1(config-vlan)#exit
Switch_Ltl1(config)#exit
Switch_Ltl1#
%SYS-5-CONFIG_I: Configured from console by console

```

Setelah pemberian nama VLAN, kemudian setting setiap interface seperti tampilan dibawah ini:

```

Switch_Ltl1#configure terminal

```

```
Enter configuration commands, one per line. End with CNTL/Z.
Switch_Lt1(config)#interface fastEthernet 0/1
Switch_Lt1(config-if)#switchport tru
Switch_Lt1(config-if)#switchport mode trunk

Switch_Lt1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed
state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed
state to up

Switch_Lt1(config-if)#exit
Switch_Lt1(config)#interface fastEthernet 0/2
Switch_Lt1(config-if)#switchport mode trunk

Switch_Lt1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed
state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed
state to up

Switch_Lt1(config-if)#switchport trunk allowed vlan 10-30
Switch_Lt1(config-if)#exit
Switch_Lt1(config)#interface fastEthernet 0/3
Switch_Lt1(config-if)#switchport mode access
Switch_Lt1(config-if)#switchport access vlan 10
Switch_Lt1(config-if)#exit
Switch_Lt1(config)#
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed
state to down

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed
state to up

Switch_Lt1(config)#interface fastEthernet 0/4
Switch_Lt1(config-if)#switchport mode access
Switch_Lt1(config-if)#switchport access vlan 20
Switch_Lt1(config-if)#exit
Switch_Lt1(config)#interface fastEthernet 0/5
Switch_Lt1(config-if)#switchport mode access
Switch_Lt1(config-if)#switchport access vlan 30
Switch_Lt1(config-if)#exit
Switch_Lt1(config)#
```

b. Konfigurasi Switch Lantai 2

Konfigurasi Router menggunakan IOS Command Line Interface (CLI). Untuk konfigurasi disini kita akan membuat switch Lt 2 kemudian konfigurasi Switch Lt 1 FastEthernet 0/1, FastEthernet 0/2, fastEthernet 0/3, fastEthernet 0/4 dan fastEthernet 0/5 untuk trunk dan menggunakan VLAN. Selanjutnya menetapkan VLAN ke port aktif pada switch Lt 2. Untuk menyetingnya dapat dilihat pada tampilan dibawah:

```
Switch>enab
Switch>enable
Switch#conf
```



```
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#host
Switch(config)#hostname Switch_Lt2
Switch_Lt2(config)#int
Switch_Lt2(config)#interface fast
Switch_Lt2(config)#interface fastEthernet 0/1
Switch_Lt2(config-if)#swi
Switch_Lt2(config-if)#switchport mode tru
Switch_Lt2(config-if)#switchport mode trunk
Switch_Lt2(config-if)#switchport tru
Switch_Lt2(config-if)#switchport trunk all
Switch_Lt2(config-if)#switchport trunk allowed vlan 10-30
Switch_Lt2(config-if)#exit
Switch_Lt2(config)#interf
Switch_Lt2(config)#interface fast
Switch_Lt2(config)#vlan 10
Switch_Lt2(config-vlan)#vlan 20
Switch_Lt2(config-vlan)#vlan 30
Switch_Lt2(config-vlan)#exit
Switch_Lt2(config)#inter
Switch_Lt2(config)#interface fas
Switch_Lt2(config)#interface fastEthernet 0/2
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state
to down

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state
to up

Switch_Lt2(config-if)#interface fastEthernet 0/2
Switch_Lt2(config-if)#swi
Switch_Lt2(config-if)#switchport mode tru
Switch_Lt2(config-if)#switchport mode trunk

Switch_Lt2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state
to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state
to up

Switch_Lt2(config-if)#interface fastEthernet 0/2
Switch_Lt2(config-if)#switchport mode trunk
Switch_Lt2(config-if)#switchport trunk all
Switch_Lt2(config-if)#switchport trunk allowed vla
Switch_Lt2(config-if)#switchport trunk allowed vlan 10-30
Switch_Lt2(config-if)#exit
Switch_Lt2(config)#interface fastEthernet 0/3
Switch_Lt2(config-if)#switchport mode access
Switch_Lt2(config-if)#switchport access vlan 10
Switch_Lt2(config-if)#exit
Switch_Lt2(config)#interface fastEthernet 0/4
Switch_Lt2(config-if)#switchport mode access
Switch_Lt2(config-if)#switchport access vlan 20
Switch_Lt2(config-if)#exit
Switch_Lt2(config)#interface fastEthernet 0/5
Switch_Lt2(config-if)#switchport mode access
Switch_Lt2(config-if)#switchport access vlan 30
```

```
Switch_Lt2(config-if)#exit
Switch_Lt2(config)#

Switch_Lt2 con0 is now available
Press RETURN to get started.
```

c. Konfigurasi Switch Lantai 3

Sama seperti pengaturan sebelumnya untuk konfigurasi disini kita akan membuat switch Lt 2 kemudian konfigurasi Switch Lt 1 FastEthernet 0/1, FastEthernet 0/2, dan fastEthernet 0/3 untuk trunk dan menggunakan VLAN. Selanjutnya menetapkan VLAN ke port aktif pada switch Lt 3. Untuk menyetingnya dapat dilihat pada tampilan dibawah:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#vlan 20
Switch(config-vlan)#vlan 30
Switch(config-vlan)#exit
Switch(config)#interface fastEthernet 0/3
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10-30
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport acces
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#

Switch con0 is now available

Press RETURN to get started.
```

C. Investigasi dan Analisa

Setelah konfigurasi, kita dapat mengecek hasilnya dengan mengexitkan terlebih dahulu kemudian check dengan membuat perintah show vlan, maka akan muncul seperti tampilan dibawah ini;

```
Switch_Lt1#show vlan

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11,
                                           Fa0/12
```

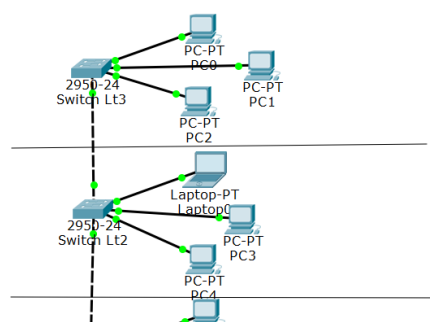
Fa0/16									Fa0/13, Fa0/14, Fa0/15,
Fa0/20									Fa0/17, Fa0/18, Fa0/19,
Fa0/24									Fa0/21, Fa0/22, Fa0/23,
10	VLAN0010							active	
20	VLAN0020							active	
30	VLAN0030							active	
1002	fddi-default							act/unsup	
1003	token-ring-default							act/unsup	
1004	fddinet-default							act/unsup	
1005	trnet-default							act/unsup	
VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1 Trans2
--									
1	enet	100001	1500	-	-	-	-	-	0 0
10	enet	100010	1500	-	-	-	-	-	0 0
20	enet	100020	1500	-	-	-	-	-	0 0

Switch_Lt1#show vlan									
VLAN	Name	Status	Ports						

1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11,						
Fa0/12									
Fa0/16			Fa0/13, Fa0/14, Fa0/15,						
Fa0/20			Fa0/17, Fa0/18, Fa0/19,						
Fa0/24			Fa0/21, Fa0/22, Fa0/23,						
10	HRD	active							
20	Finance	active							
30	Marketing	active							
1002	fddi-default	act/unsup							
1003	token-ring-default	act/unsup							
1004	fddinet-default	act/unsup							
1005	trnet-default	act/unsup							
VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1 Trans2

1	enet	100001	1500	-	-	-	-	-	0 0
10	enet	100010	1500	-	-	-	-	-	0 0
20	enet	100020	1500	-	-	-	-	-	0 0

1. Konfigurasi Switch Lantai 1



View Tools Ext

Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	--	--	0030.A3AB.1301
FastEthernet0/2	Up	--	--	0030.A3AB.1302
FastEthernet0/3	Up	10	--	0030.A3AB.1303
FastEthernet0/4	Up	20	--	0030.A3AB.1304
FastEthernet0/5	Up	30	--	0030.A3AB.1305
FastEthernet0/6	Down	1	--	0030.A3AB.1306
FastEthernet0/7	Down	1	--	0030.A3AB.1307
FastEthernet0/8	Down	1	--	0030.A3AB.1308
FastEthernet0/9	Down	1	--	0030.A3AB.1309
FastEthernet0/10	Down	1	--	0030.A3AB.130A
FastEthernet0/11	Down	1	--	0030.A3AB.130B
FastEthernet0/12	Down	1	--	0030.A3AB.130C
FastEthernet0/13	Down	1	--	0030.A3AB.130D
FastEthernet0/14	Down	1	--	0030.A3AB.130E
FastEthernet0/15	Down	1	--	0030.A3AB.130F
FastEthernet0/16	Down	1	--	0030.A3AB.1310
FastEthernet0/17	Down	1	--	0030.A3AB.1311
FastEthernet0/18	Down	1	--	0030.A3AB.1312
FastEthernet0/19	Down	1	--	0030.A3AB.1313
FastEthernet0/20	Down	1	--	0030.A3AB.1314
FastEthernet0/21	Down	1	--	0030.A3AB.1315
FastEthernet0/22	Down	1	--	0030.A3AB.1316
FastEthernet0/23	Down	1	--	0030.A3AB.1317
FastEthernet0/24	Down	1	--	0030.A3AB.1318
Vlan1	Down	1	<not set>	0006.2A6E.63B2

Hostname: Switch_Lt1

Physical Location: Intercity, Home City, Corporate Office, Main Wiring Closet

1841 Router0 --- 2950-24 Switch Lt1 --- Fa0/24 Laptop-PT Laptop1

PC-PT PC6

Switch_Lt1#show vlan

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
10	HRD	active	Fa0/3
20	Finance	active	Fa0/4
30	Marketing	active	Fa0/5
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
30	enet	100030	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0

PC 1 Lt 1

1841 Router0 --- 2950-24 Switch Lt1 --- PC-PT PC6

PC-PT PC6

Physical Location: Intercity, Home City, Corporate Office, Wiring Closet

PC5

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static

IP Address 192.168.0.3

Subnet Mask 255.255.255.240

Default Gateway 192.168.0.1

DNS Server

IPv6 Configuration

☐ DHCP ☒ Auto Config ☒ Static

IPv6 Address

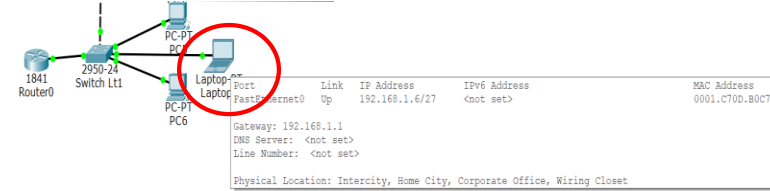
Link Local Address FE80::201:63FF:FEA3:3E8A

IPv6 Gateway

IPv6 DNS Server

Dialer Editor Firewall

PC 2 Lt 1



Laptop1

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration
☒ DHCP ☐ Static

IP Address 192.168.1.6
 Subnet Mask 255.255.255.224
 Default Gateway 192.168.1.1
 DNS Server

IPv6 Configuration
☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address
 Link Local Address FE80::201:C7FF:FE0D:B0C7
 IPv6 Gateway
 IPv6 DNS Server

PC 3 Lt 1



PC6

Physical Config Desktop Custom Interface

IP Configuration

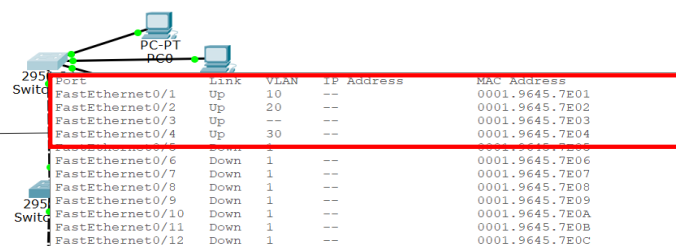
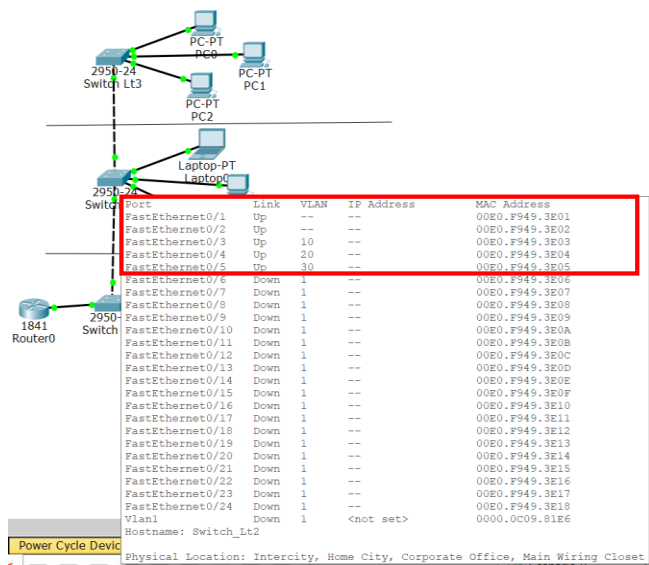
IP Configuration
☒ DHCP ☐ Static

IP Address 192.168.2.11
 Subnet Mask 255.255.255.128
 Default Gateway 192.168.2.1
 DNS Server

IPv6 Configuration
☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address
 Link Local Address FE80::260:5CFF:FE38:CB9B
 IPv6 Gateway
 IPv6 DNS Server

2. konfigurasi switch lantai 2

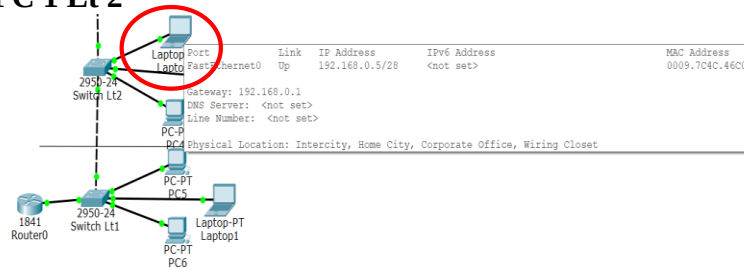


Switch_Lt2#show vlan

VLAN Name	Status	Ports
1 default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
10 VLAN0010	active	Fa0/3
20 VLAN0020	active	Fa0/4
30 VLAN0030	active	Fa0/5
1002 fddi-default	act/unsup	
1003 token-ring-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trnet-default	act/unsup	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
30	enet	100030	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0

PC 1 Lt 2



Laptop0

Physical Config Desktop Custom Interface

IP Configuration

☒ DHCP ☐ Static

IP Address 192.168.0.5

Subnet Mask 255.255.255.240

Default Gateway 192.168.0.1

DNS Server

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

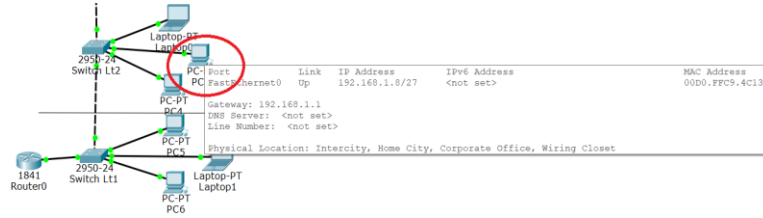
IPv6 Address

Link Local Address FE80::209:7CFF:FE4C:46C0

IPv6 Gateway

IPv6 DNS Server

PC 2 Lt 2



PC3

Physical Config Desktop Custom Interface

IP Configuration

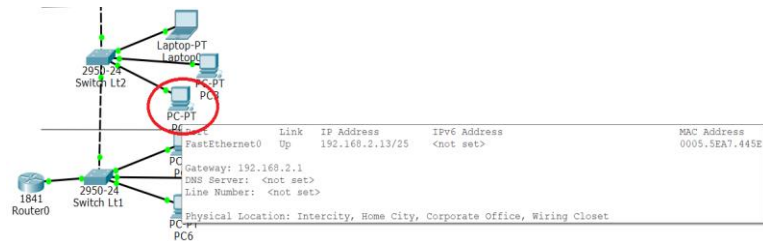
IP Configuration
☒ DHCP ☐ Static

IP Address 192.168.1.8
 Subnet Mask 255.255.255.224
 Default Gateway 192.168.1.1
 DNS Server

IPv6 Configuration
☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address
 Link Local Address FE80::2D0:FFFF:FEC9:4C13
 IPv6 Gateway
 IPv6 DNS Server

PC 3 Lt 2



PC4

Physical Config Desktop Custom Interface

IP Configuration

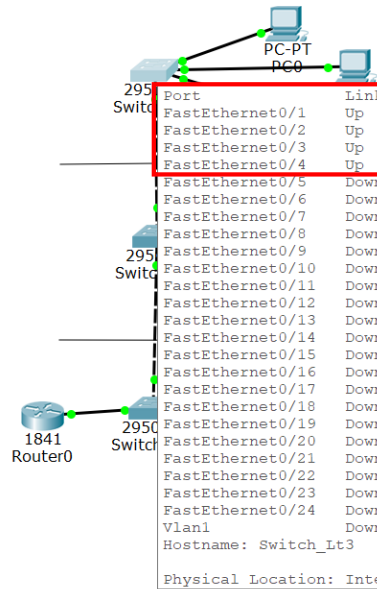
IP Configuration
☒ DHCP ☐ Static

IP Address 192.168.2.13
 Subnet Mask 255.255.255.128
 Default Gateway 192.168.2.1
 DNS Server

IPv6 Configuration
☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address
 Link Local Address FE80::205:5EFF:FEA7:445E
 IPv6 Gateway
 IPv6 DNS Server

3. Konfigurasi Switch Lantai 3



Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	10	--	0001.9645.7E01
FastEthernet0/2	Up	20	--	0001.9645.7E02
FastEthernet0/3	Up	--	--	0001.9645.7E03
FastEthernet0/4	Up	30	--	0001.9645.7E04
FastEthernet0/5	Down	1	--	0001.9645.7E05
FastEthernet0/6	Down	1	--	0001.9645.7E06
FastEthernet0/7	Down	1	--	0001.9645.7E07
FastEthernet0/8	Down	1	--	0001.9645.7E08
FastEthernet0/9	Down	1	--	0001.9645.7E09
FastEthernet0/10	Down	1	--	0001.9645.7E0A
FastEthernet0/11	Down	1	--	0001.9645.7E0B
FastEthernet0/12	Down	1	--	0001.9645.7E0C
FastEthernet0/13	Down	1	--	0001.9645.7E0D
FastEthernet0/14	Down	1	--	0001.9645.7E0E
FastEthernet0/15	Down	1	--	0001.9645.7E0F
FastEthernet0/16	Down	1	--	0001.9645.7E10
FastEthernet0/17	Down	1	--	0001.9645.7E11
FastEthernet0/18	Down	1	--	0001.9645.7E12
FastEthernet0/19	Down	1	--	0001.9645.7E13
FastEthernet0/20	Down	1	--	0001.9645.7E14
FastEthernet0/21	Down	1	--	0001.9645.7E15
FastEthernet0/22	Down	1	--	0001.9645.7E16
FastEthernet0/23	Down	1	--	0001.9645.7E17
FastEthernet0/24	Down	1	--	0001.9645.7E18
Vlan1	Down	1	<not set>	0007.EC0E.5706

Hostname: Switch_Lt3

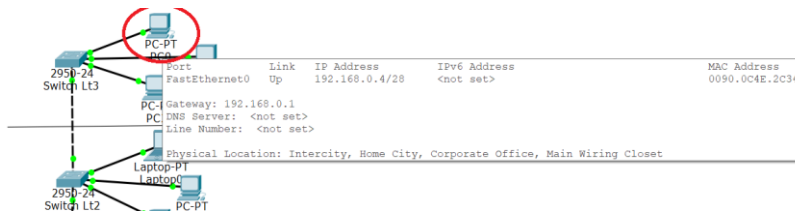
Physical Location: Intercity, Home City, Corporate Office, Main Wiring Closet

Switch_Lt3#show vlan

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
10	VLAN0010	active	Fa0/1
20	VLAN0020	active	Fa0/2
30	VLAN0030	active	Fa0/4
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
30	enet	100030	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0

PC 1 Lt 3



PC0

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static

IP Address 192.168.0.4

Subnet Mask 255.255.255.240

Default Gateway 192.168.0.1

DNS Server

IPv6 Configuration

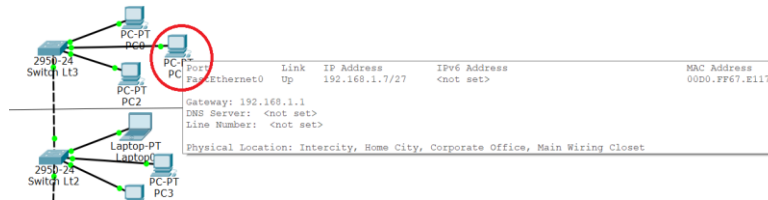
☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address

Link Local Address FE80::290:CFF:FE4E:2C34

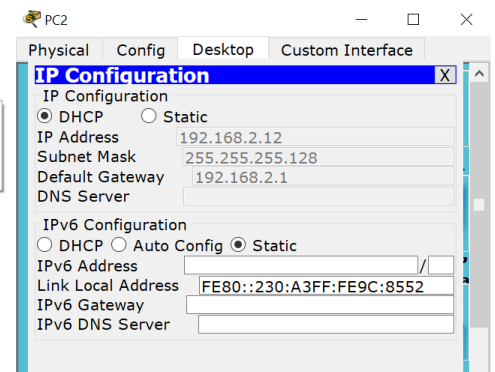
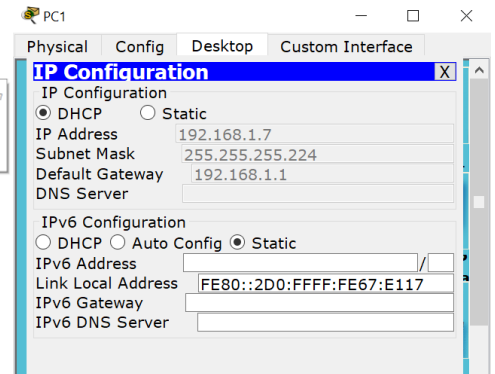
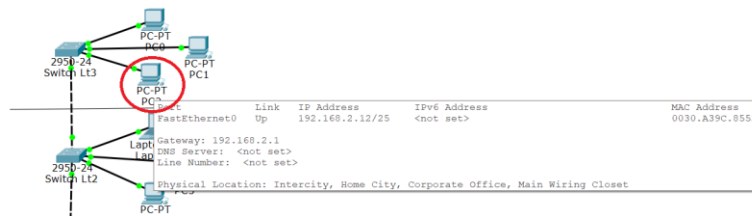
IPv6 Gateway

IPv6 DNS Server



PC 2 Lt 3

PC 3 Lt 3



D. Cek Koneksi

Jika konfigurasi telah selesai dilakukan, untuk melakukan pengecekan apakah konfigurasi sudah berhasil atau tidak dengan melakukan ping dari tiap PC . Uji koneksi tiap PC dengan cara klik kiri PC1 → Tab Desktop → Pilih Command Prompt → Untuk mengecek apakah Simulasi Jaringan VLAN yang tadi sudah dibuat berjalan dengan baik atau tidak dapat dilakukang dengan cara PING. Jika muncul tulisan Reply from 192.168.1.6: bytes=32 time=1ms TTL=127 maka laptop telah terhubung dan terkoneksi dengan baik.

✓ Antar PC dalam Satu Lantai

Ping dari PC 1 ke PC 2 di Lantai 1

```
PC>ping 192.168.1.6

Pinging 192.168.1.6 with 32 bytes of data:

Reply from 192.168.1.6: bytes=32 time=0ms TTL=127
Reply from 192.168.1.6: bytes=32 time=0ms TTL=127
Reply from 192.168.1.6: bytes=32 time=1ms TTL=127
Reply from 192.168.1.6: bytes=32 time=0ms TTL=127

Ping statistics for 192.168.1.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping dari PC 1 ke PC 3 di Lantai 1

```
PC>ping 192.168.2.11

Pinging 192.168.2.11 with 32 bytes of data:

Reply from 192.168.2.11: bytes=32 time=0ms TTL=127
Reply from 192.168.2.11: bytes=32 time=0ms TTL=127
Reply from 192.168.2.11: bytes=32 time=1ms TTL=127
Reply from 192.168.2.11: bytes=32 time=0ms TTL=127

Ping statistics for 192.168.2.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping dari PC 1 ke PC 2 di Lantai 2

```
PC>ping 192.168.1.8

Pinging 192.168.1.8 with 32 bytes of data:

Reply from 192.168.1.8: bytes=32 time=10ms TTL=127
Reply from 192.168.1.8: bytes=32 time=0ms TTL=127
Reply from 192.168.1.8: bytes=32 time=0ms TTL=127
Reply from 192.168.1.8: bytes=32 time=0ms TTL=127

Ping statistics for 192.168.1.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

Ping dari PC 1 ke PC 3 di Lantai 2

```
PC>ping 192.168.2.13

Pinging 192.168.2.13 with 32 bytes of data:

Reply from 192.168.2.13: bytes=32 time=1ms TTL=127
Reply from 192.168.2.13: bytes=32 time=0ms TTL=127
Reply from 192.168.2.13: bytes=32 time=0ms TTL=127
Reply from 192.168.2.13: bytes=32 time=0ms TTL=127

Ping statistics for 192.168.2.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping dari PC 1 ke PC 2 di Lantai 3

```
PC>ping 192.168.1.7

Pinging 192.168.1.7 with 32 bytes of data:

Reply from 192.168.1.7: bytes=32 time=0ms TTL=127
Reply from 192.168.1.7: bytes=32 time=0ms TTL=127
Reply from 192.168.1.7: bytes=32 time=1ms TTL=127
Reply from 192.168.1.7: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.1.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping dari PC 1 ke PC 3 di Lantai 3

```
PC>ping 192.168.2.12

Pinging 192.168.2.12 with 32 bytes of data:

Reply from 192.168.2.12: bytes=32 time=1ms TTL=127
Reply from 192.168.2.12: bytes=32 time=0ms TTL=127
Reply from 192.168.2.12: bytes=32 time=1ms TTL=127
Reply from 192.168.2.12: bytes=32 time=0ms TTL=127

Ping statistics for 192.168.2.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

✓ Antar PC dalam Satu VLAN

Ping antar VLAN 10

```
PC>ping 192.168.0.4

Pinging 192.168.0.4 with 32 bytes of data:

Reply from 192.168.0.4: bytes=32 time=54ms TTL=128
Reply from 192.168.0.4: bytes=32 time=0ms TTL=128
Reply from 192.168.0.4: bytes=32 time=0ms TTL=128
Reply from 192.168.0.4: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 54ms, Average = 13ms
```

```
PC>ping 192.168.0.3

Pinging 192.168.0.3 with 32 bytes of data:

Reply from 192.168.0.3: bytes=32 time=0ms TTL=128
Reply from 192.168.0.3: bytes=32 time=0ms TTL=128
Reply from 192.168.0.3: bytes=32 time=0ms TTL=128
Reply from 192.168.0.3: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping antar VLAN 20

```
PC>ping 192.168.1.8

Pinging 192.168.1.8 with 32 bytes of data:

Reply from 192.168.1.8: bytes=32 time=0ms TTL=128
Reply from 192.168.1.8: bytes=32 time=0ms TTL=128
Reply from 192.168.1.8: bytes=32 time=6ms TTL=128
Reply from 192.168.1.8: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.1.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms

PC>ping 192.168.1.7

Pinging 192.168.1.7 with 32 bytes of data:

Reply from 192.168.1.7: bytes=32 time=0ms TTL=128
Reply from 192.168.1.7: bytes=32 time=2ms TTL=128
Reply from 192.168.1.7: bytes=32 time=0ms TTL=128
Reply from 192.168.1.7: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.1.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Ping antar VLAN 30

```
PC>ping 192.168.2.13

Pinging 192.168.2.13 with 32 bytes of data:

Reply from 192.168.2.13: bytes=32 time=0ms TTL=128
Reply from 192.168.2.13: bytes=32 time=1ms TTL=128
Reply from 192.168.2.13: bytes=32 time=0ms TTL=128
Reply from 192.168.2.13: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.2.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>ping 192.168.2.12

Pinging 192.168.2.12 with 32 bytes of data:

Reply from 192.168.2.12: bytes=32 time=1ms TTL=128
Reply from 192.168.2.12: bytes=32 time=0ms TTL=128
Reply from 192.168.2.12: bytes=32 time=0ms TTL=128
Reply from 192.168.2.12: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.2.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

✓ Antar PC Beda VLAN (Beda Departemen) Beda Lantai

```
PC>ping 192.168.1.8

Pinging 192.168.1.8 with 32 bytes of data:

Reply from 192.168.1.8: bytes=32 time=0ms TTL=127
Reply from 192.168.1.8: bytes=32 time=1ms TTL=127
Reply from 192.168.1.8: bytes=32 time=0ms TTL=127
Reply from 192.168.1.8: bytes=32 time=6ms TTL=127

Ping statistics for 192.168.1.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms

PC>ping 192.168.2.13

Pinging 192.168.2.13 with 32 bytes of data:

Reply from 192.168.2.13: bytes=32 time=0ms TTL=127
Reply from 192.168.2.13: bytes=32 time=0ms TTL=127
Reply from 192.168.2.13: bytes=32 time=1ms TTL=127
Reply from 192.168.2.13: bytes=32 time=0ms TTL=127

Ping statistics for 192.168.2.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
PC>ping 192.168.1.7

Pinging 192.168.1.7 with 32 bytes of data:

Reply from 192.168.1.7: bytes=32 time=0ms TTL=127
Reply from 192.168.1.7: bytes=32 time=0ms TTL=127
Reply from 192.168.1.7: bytes=32 time=6ms TTL=127
Reply from 192.168.1.7: bytes=32 time=2ms TTL=127

Ping statistics for 192.168.1.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 2ms

PC>ping 192.168.2.12

Pinging 192.168.2.12 with 32 bytes of data:

Reply from 192.168.2.12: bytes=32 time=1ms TTL=127
Reply from 192.168.2.12: bytes=32 time=0ms TTL=127
Reply from 192.168.2.12: bytes=32 time=1ms TTL=127
Reply from 192.168.2.12: bytes=32 time=7ms TTL=127

Ping statistics for 192.168.2.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 2ms
```

Pada percobaan ini, simulasi jaringan dengan satu router, 3 switch dimana setiap switch dipasang 3 PC/laptop sudah dibuat dan berjalan dengan baik dengan menggabungkan semua port pada switch yang digunakan untuk komputer menjadi satu jalur dan seolah-olah dalam satu ruang yang sama dengan memberikan vlan yang sama

pada setiap portnya. Hal tersebut mengakibatkan 3 komputer yang terhubung ke switch dapat saling terkoneksi satu sama lain dengan ip address yang masih dalam satu jaringan. Dibuktikan dengan menggunakan perintah “PING” ke ip address komputer masing-masing atau dengan menggunakan fitur bertanda surat di Cisco Packet Tracer dan akan muncul status “Successful” atau “Reply Form....”. Semua device telah terhubung (penghubung antar router) dan straight (penghubung antara router dengan switch) yang tadinya bertanda merah berubah menjadi hijau yang menandakan tanda sudah saling terhubung. Kemudian berbagai pengujian tes ping sudah dilakukan menghasilkan semua device yang sudah dapat saling terhubung dengan baik. Loss = 0% menandakan proses pengiriman paket ping dari setiap PC berjalan sukses tanpa error.

E. KESIMPULAN

Melakukan simulasi LAN sangat bermanfaat sebelum menerapkan jaringan tersebut secara nyata. Diharapkan dapat mencegah *error* saat men-*setting* LAN secara nyata dan tentunya menghemat waktu instalasi jaringan. Penggunaan VLAN sangat bermanfaat sekali untuk kemajuan pembagian Network agar lebih efisien. VLAN merupakan suatu model jaringan yang tidak terbatas pada lokasi fisik seperti LAN sehingga mengakibatkan suatu network dapat dikonfigurasi secara virtual tanpa harus menuruti lokasi fisik peralatan. Penggunaan VLAN akan membuat pengaturan jaringan menjadi sangat fleksibel karena dapat dibuat segmen yang bergantung pada organisasi, tanpa bergantung lokasi workstations.